FIFA WORLD CUP AI PREDICTOR

End-to-End Data Science Project

Pablo | Data Science Bootcamp 2025

PROJECT OVERVIEW

Goal

Predict international football match outcomes using
150+ years of historical data, 4 ML/DL models, and Generative AI — deployed as an interactive app.

Pipeline

- 1. Data Collection — 49K matches + FIFA rankings
- 2. Data Cleaning — Handle nulls, parse dates, classify results
- 3. Feature Engineering — 13 contextual features per match
- 4. EDA — 7+ visualizations across multiple dimensions
- 5. ML/DL — XGBoost, Random Forest, Gradient Boosting, MLP
- 6. GenAI — LLM integration via OpenRouter (RAG-style)
- 7. Deployment — Streamlit app with 5 interactive pages

DATA SOURCES

results.csv

49,287 matches

International football 1872–2024

9 columns: date, teams, scores, tournament, city, country, neutral

ranking.csv

67,472 records

FIFA World Rankings 1992–2024

8 columns: rank, country, points, confederation, date

WC 2026 fixture

104 matches

Official FIFA schedule

48 teams, 12 groups, 16 stadiums

Group + full knockout bracket

FEATURE ENGINEERING

13 contextual features computed per match — using only pre-match data (no leakage)

Category	Features	Description
Rankings	home_ranking, away_ranking, ranking_diff	FIFA ranking at match date + gap
Form	home_form, away_form, form_diff	Win % in last 10 matches + gap
Goals	home_goals_avg, away_goals_avg	Avg goals scored (last 10 matches)
H2H	h2h_hw_rate, h2h_draw_rate, h2h_aw_rate, h2h_total	Historical head-to-head rates + count
Venue	is_neutral	Neutral ground flag

EXPLORATORY DATA ANALYSIS

42.4%

Home Win Rate
in World Cups

37.0%

Away Win Rate
in World Cups

20.7%

Draw Rate
in World Cups

- Brazil leads all-time World Cup wins (76), followed by Germany (68) and Argentina (47)
- International football has grown exponentially: 15,000+ matches in the 2010s vs ~300 in the 1870s
- The most predictive features: ranking difference, H2H win rates, and recent form
- 7 visualizations created: pie charts, bar charts, heatmaps, choropleth map, correlation plots

MACHINE LEARNING MODELS

Model	Type	Test Acc	CV Acc (5K)	Key Strength
⭐ **XGBoost**	Gradient Boosting	**~62%**	~62%	Best overall performance
Random Forest	Bagging Ensemble	~61%	~61%	Robust, less overfitting
Gradient Boosting	Sklearn Boosting	~61%	~61%	Good baseline reference
Neural Network	MLP (Keras)	~49%	—	Non-linear patterns

Note: Football match prediction has a natural accuracy ceiling of ~65–70% due to the sport's inherent randomness. Our ~62% accuracy is consistent with published academic research on football prediction models.

GENERATIVE AI INTEGRATION

LLM Integration

OpenRouter API (free tier)
6-model fallback system
Llama 3.3, Nemotron,
Gemma auto-rotation

Data-Grounded

Real statistics passed
as context to the LLM
FIFA rankings, H2H records,
form data from dataset

RAG Chatbot

Interactive Q&A about
World Cup history
Answers grounded in
49K+ match records

STREAMLIT APPLICATION

5 interactive pages — live demo follows this slide

📊 **Dashboard**	7 EDA visualizations, H2H explorer, world map
🔮 **Predictor**	Pick any 2 teams ⟶ XGBoost prediction with probabilities
🏆 **WC26 Predictor**	Full World Cup 2026 simulation: groups + knockout bracket
🤖 **AI Analyst**	LLM-generated match analysis in English or Spanish
💬 **Chat with Data**	RAG chatbot answering World Cup history questions

KEY TAKEAWAYS

Conclusions

- XGBoost achieved best accuracy (~62%), consistent with research
- Ranking difference and H2H records are the strongest predictors
- Home advantage adds ~5% win probability in World Cup matches
- Draws are hardest to predict (low recall across all models)
- Data-grounded GenAI produces factual, useful sports analysis

Future Work

- Add player-level data (injuries, key absences)
- Include betting odds as market-implied features
- Implement proper RAG with vector embeddings
- Add LSTM time-series models for sequence effects
- Deploy to cloud (Streamlit Community Cloud)

Thank you! — Questions?